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Tuberculosis in the 21st Century: Epidemiology, Pathophysiology, Risk Assessment

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Session Objectives

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- Explain the epidemiology, pathogenesis and transmission of TB
- Identify risk factors associated with TB
- Understand patient presentation and physical examination findings consistent with active TB
- Differentiate the radiologic findings of TB
- Differentiate between testing modalities for screening and diagnosis of active vs. latent TB

Source: Course Syllabus

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COMLEX-USA
National Board of Osteopathic Medical Examiners

9. Patient Presentations Related to the Respiratory System

9.1 Cough

9.9 Respiratory Chest Pain

• respiratory tract infections (**tuberculosis**)

CONSTITUTIONAL SIGNS AND SYMPTOMS

fatigue • fever • generalized weakness • involuntary weight loss • malaise • night sweats • pallor

PHYSICAL EXAM FINDINGS

pulmonary auscultation, percussion, and palpation (eg, egophony, stridor, rales, rhonchus, wheeze, bronchophony, fremitus, chest wall expansion, diaphragmatic excursion)

LABORATORY TEST FINDINGS AND DIAGNOSTIC IMAGING

• CT scanning • evaluation of pleural effusions (eg, exudate, transudate) • fluoroscopy • interpretation of pulse oximetry • • purified protein derivative (PPD) skin test • radiography • sputum Gram stain, culture and sensitivity

USMLE United States Medical Licensing Examination

- **Respiratory System**
- Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis
- Infectious, immunologic, and inflammatory disorders
  - **pulmonary tuberculosis**

https://www.nbome.org/assessments/comlex-usa/master-blueprint/clinical-presentations/circulatory-and-hematologic-systems/
https://www.usmle.org/prepare-your-exam/step-1-materials/step-1-content-outline-and-specifications

Definitions

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Infection: Presence of bacteria in the body which may be symptomatic (active) or not symptomatic (latent infection)

Latent TB Infection (LTBI): TB bacteria lives in body, but does not cause symptoms

- Many people who have LTBI may not develop TB disease
- People do not feel sick & cannot spread TB to other people

TB Disease: When TB bacteria becomes active and causes symptoms

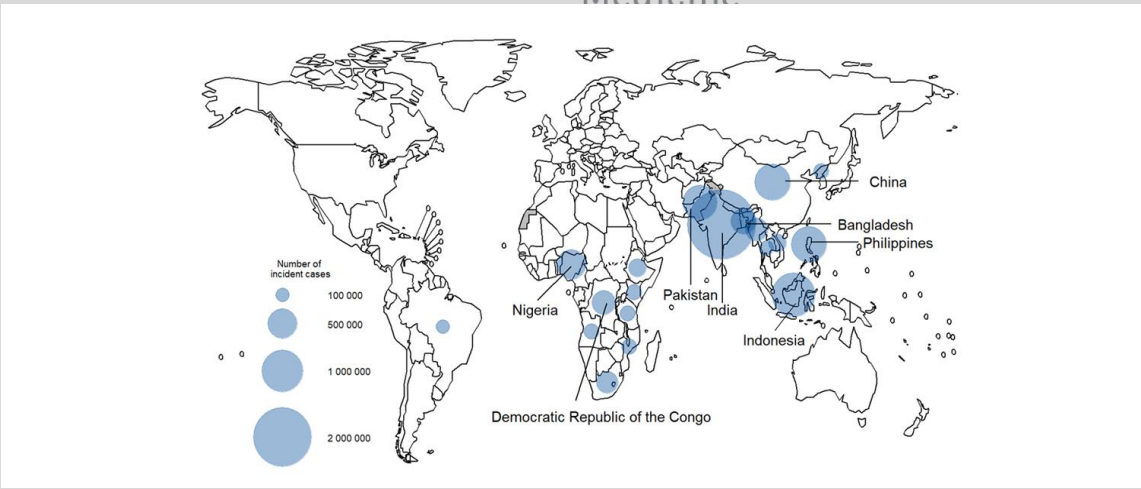
Post-primary disease = reactivation TB = secondary TB

Source: https://www.cdc.gov/tb/about/index.html

Epidemiology: Worldwide

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- 2 billion people infected with TB bacteria worldwide
- 2024: 1.23 million deaths
- 2024: TB resurged as #1 Infectious disease killer
- #1 cause of death for people living with HIV
- TB ANYWHERE IS TB EVERYWHERE

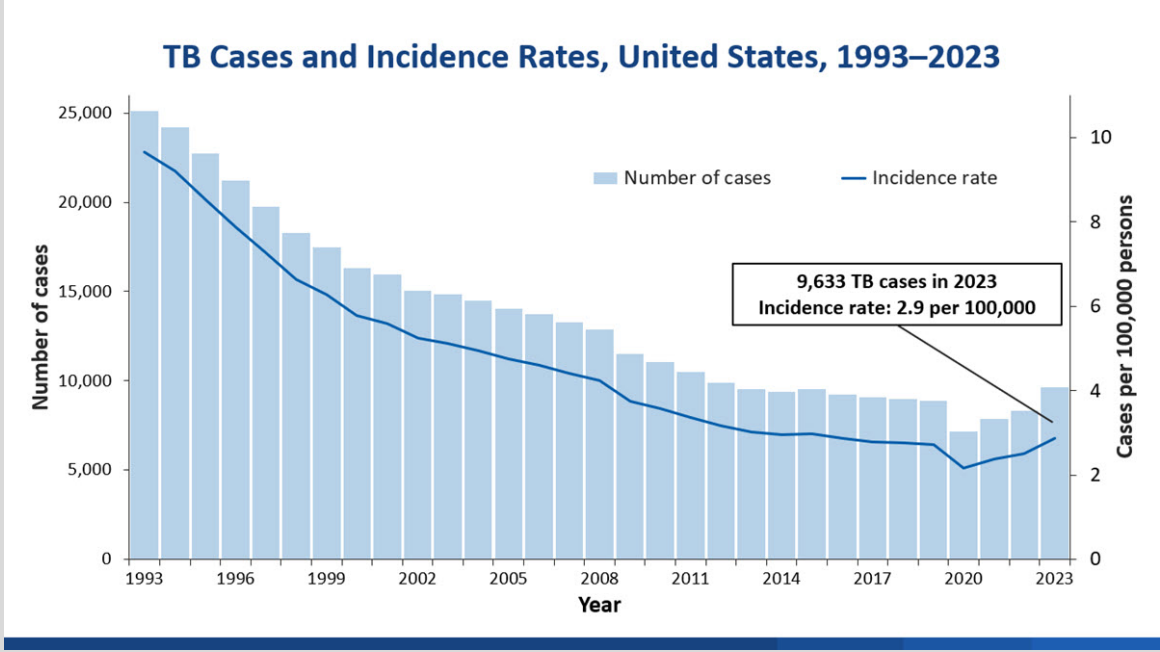


Source: https://www.cdc.gov/global-hiv-tb/php/globaltb/index.html

Epidemiology: USA

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- 10,260: reported TB cases in 2025 (3 cases/100,000 persons)
- 13 million people living with LTBI
- Disproportionally affects non-U.S.–born and racial and ethnic minority groups
- TB cases and rate increased since 2021:
  - post-pandemic travel and migration;
  - outbreaks in several states
- 2024-2025: decrease of 1% in cases and 2% in rates



https://www.cdc.gov/tb-data/2024-provisional/index.html

History

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- 20,000-15,000 years ago: common ancestor of *M. tuberculosis* appeared
- 2400 B.C.E: Egyptian mummies show signs of TB-associated skeletal deformities
- Ancient Greece: “Phthisis,” Hippocrates believed disease was hereditary
- Middle Ages: “King’s Evil,” believed to be healed after a royal touch
- 1800’s: Consumption
- 1882: Robert Koch discovered TB caused by infectious agent



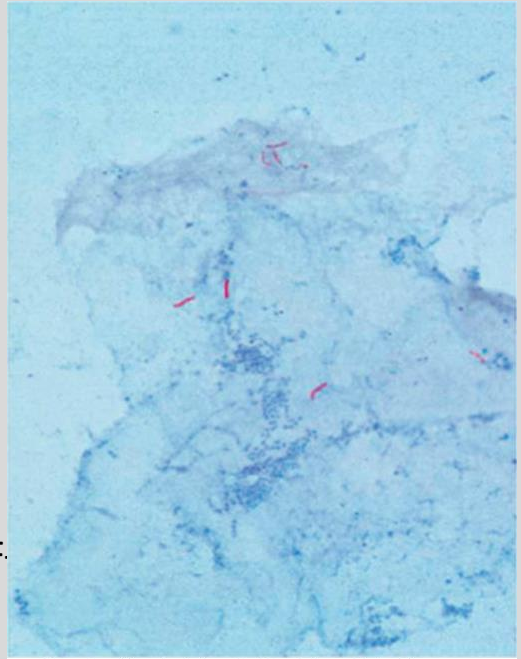
Source: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5432783/
Image: Saranac Lake Free Library
https://www.pbs.org/wgbh/americanexperience/features/plague-gallery/

Mycobacterium tuberculosis

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- Rod-shaped, non-spore forming, aerobic bacteria
- Facultative intracellular
- Acid Fast → high content of mycolic acid
  - Mycolic acid: long-chain cross linked fatty acid
- Neutral on gram stain
  - Mycolic acid makes impervious to staining
- Slow growing
- Virulence factors:
  - Cord factor: activates macrophages and induces release of TNF-alpha
  - Sulfatides: inhibit phagolysosome fusion



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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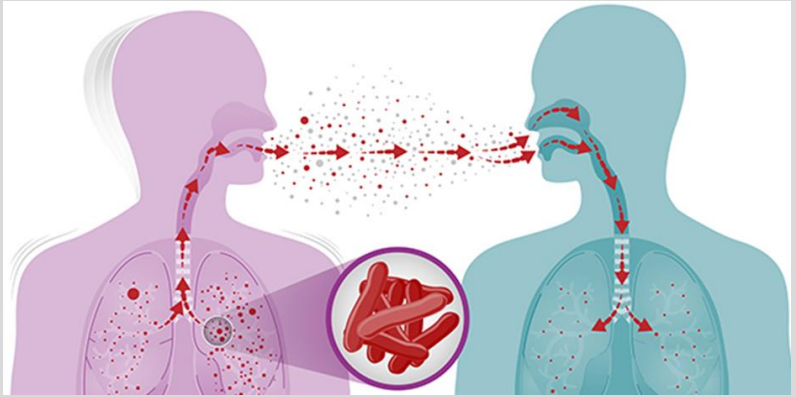
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Transmission

- Carried on airborne particles called droplet nuclei (1-5 microns)
- Aerosolized by coughing, sneezing and speaking
  - Ex. 3,000 infectious nuclei per cough
- Can remain suspended in air for several hrs
- Not transmitted via surface contact
- Likelihood of transmission dependent on:
  - Probability of contact
  - Intimacy & duration of contact
  - Degree of infectiousness
  - Shared environment



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https://www.cdc.gov/tb/index.html

Risk Factors

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Two General Categories

1. Persons who have been recently infected with TB bacteria
   - Close contact to a person with infectious disease
   - Immigrated from area with high rates of TB
   - Children <5yo who have positive TB test
   - Groups with high rates of transmission: homeless, injection drug users, HIV+
   - Work or reside with people who are high risk in facilities or institutions: hospitals, homeless shelters, correctional facilities, nursing homes & residential homes for those with HIV
2. Persons with medical conditions that weaken the immune system

Risk Factors for Active Tuberculosis in Persons Who Have Been Infected with Tubercle Bacilli

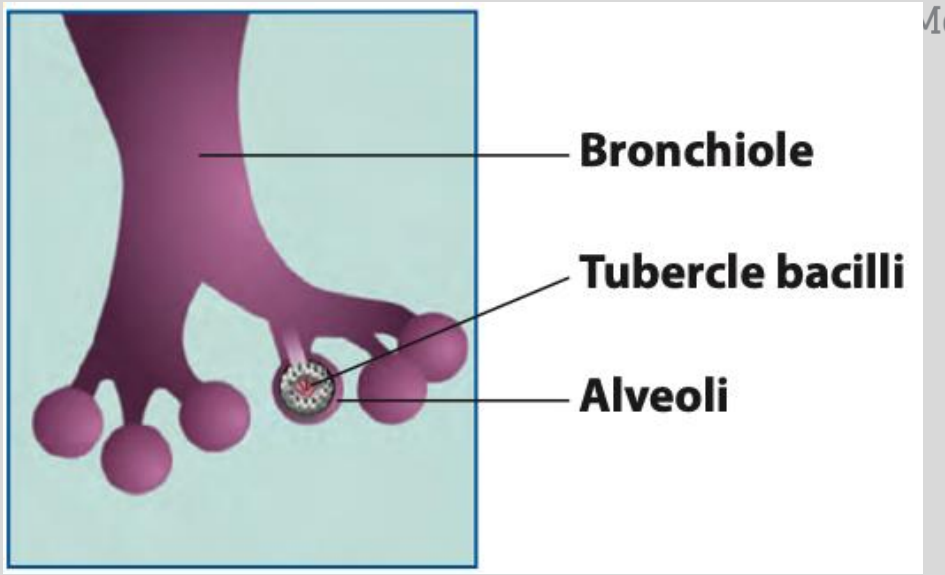
FACTOR	RELATIVE RISK/ODDS[a]
Recent infection (<1 year)	12.9
Fibrotic lesions (spontaneously healed)	2–20
Comorbidities and iatrogenic causes	
HIV infection	21–>30
Silicosis	30
Chronic renal failure/hemodialysis	10–25
Diabetes	2–4
IV drug use	10–30
Excessive alcohol use	3
Immunosuppressive treatment	10
Tumor necrosis factor α inhibitors	4–5
Gastrectomy	2–5
Jejunoileal bypass	30–60
Posttransplantation period (renal, cardiac)	20–70
Tobacco smoking	2–3
Malnutrition and severe underweight	2

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https://www.cdc.gov/tb/index.html

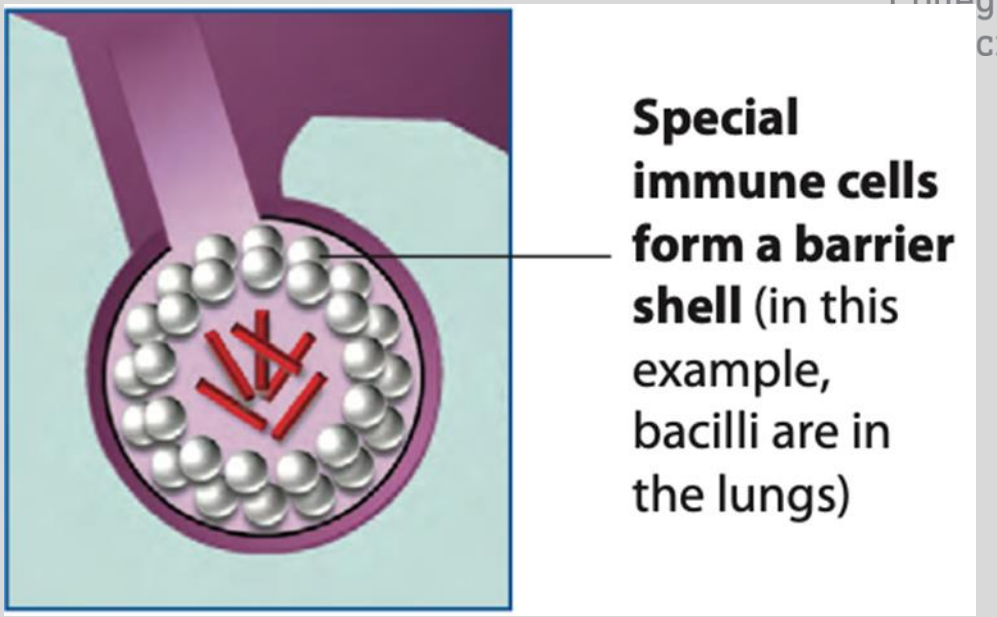
Pathogenesis

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Source: https://www.cdc.gov/tb/index.html

Pathogenesis

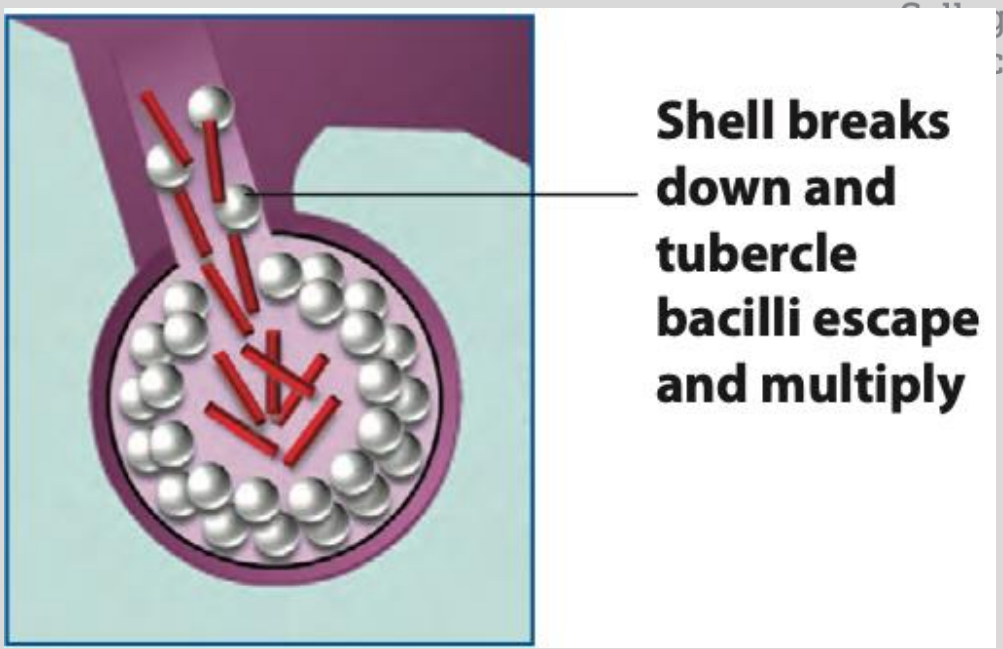


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Pathogenesis

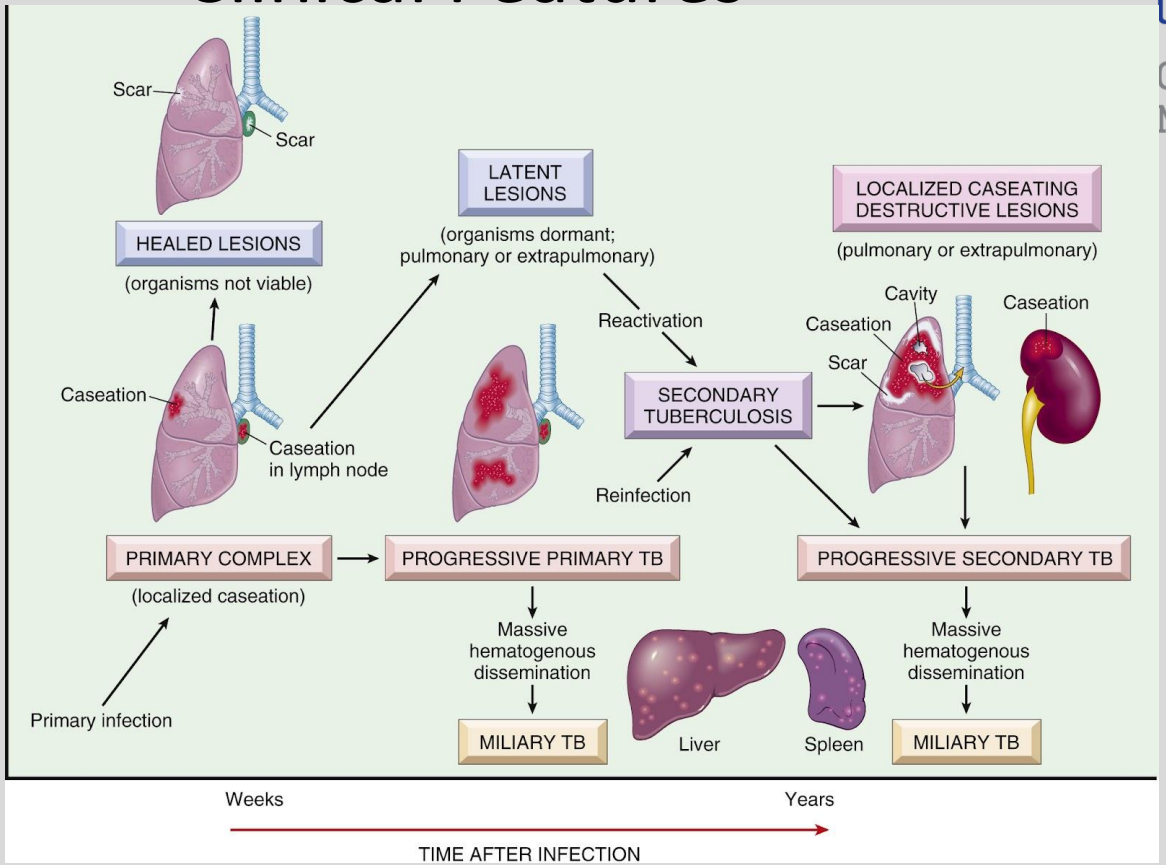
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Source: https://www.cdc.gov/tb/index.html

Clinical Features



Source: Source: Robbins & Cotran Pathologic Basis of Disease, Chapter 8 Infectious Diseases.

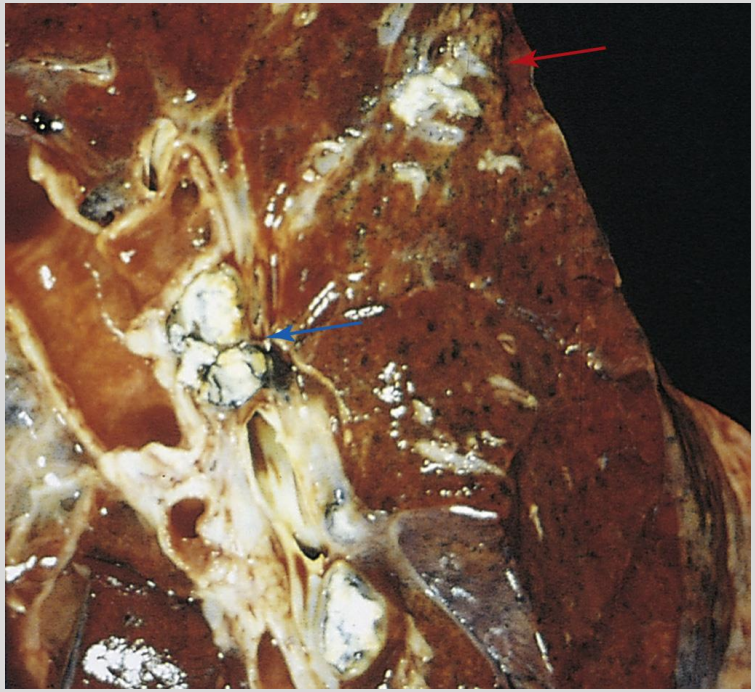
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Histopathology

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- Primary:
  - Ghon complex:
    - Ghon focus: 1-1.5 cm area of gray-white inflammation with consolidation (red arrow)
    - Middle and lower lung zones
    - Hilar Lymph Nodes with caseation (Blue arrow)
  - Ghon complex undergoes fibrosis → calcification → Ranke complex
  - Histology: Central caseation surrounded by epithelioid and multinucleated giant cells

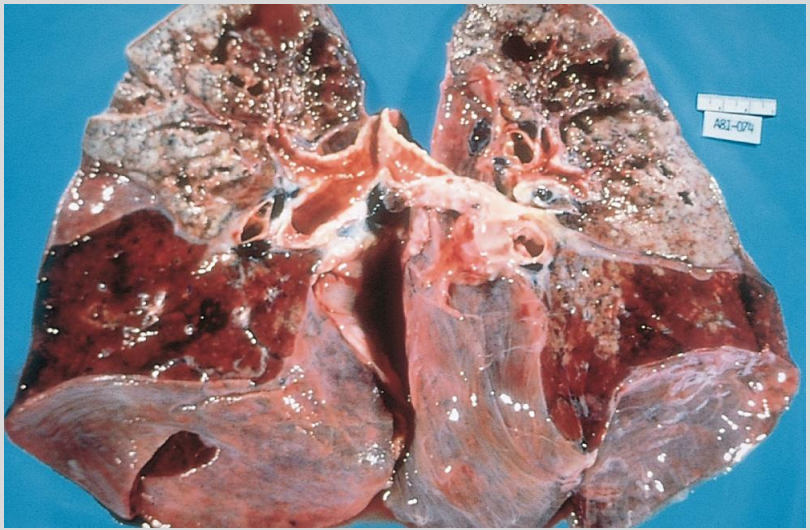


Source: Robbins & Cotran Pathologic Basis of Disease, Chapter 8 Infectious Diseases.

Histopathology

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- Secondary:
  - Lesion: small focus of consolidation, <2cm diameter, within 1-2cm of apical lung
  - Initial parenchymal focus undergoes progressive fibrous encapsulation → fibrocalcific scars
  - Histology: coalescent tubercles with central caseation



Source: Robbins & Cotran Pathologic Basis of Disease, Chapter 8 Infectious Diseases

Clinical Features

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Woman with LTBI

Person with LTBI (Infected)	Person with TB Disease (Infectious)
Has a small amount of TB bacteria in his/her body that are alive, but inactive	Has a large amount of active TB bacteria in his/her body
Cannot spread TB bacteria to others	May spread TB bacteria to others
Does **not** feel sick, but may become sick if the bacteria become active in his/her body	May feel sick and may have symptoms such as a cough, fever, and/or weight loss
Usually has a TB skin test or TB blood test reaction indicating TB infection	Usually has a TB skin test or TB blood test reaction indicating TB infection
Radiograph is typically normal	Radiograph may be abnormal
Sputum smears and cultures are negative	Sputum smears and cultures may be positive
Should consider treatment for LTBI to prevent TB disease	Needs treatment for TB disease
Does **not** require respiratory isolation	May require respiratory isolation
Not a TB case	A TB case



Woman with TB disease

Source: https://www.cdc.gov/tb/index.html

Clinical Features: Extra-Pulmonary

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- Tuberculous lymphadenitis: TB of lymph nodes
- Pleural TB: an exudative pleural effusion with predominantly lymphocytes
- Upper airway TB: larynx, pharynx and epiglottis
- Genitourinary TB: symptoms may resemble UTI
- Skeletal TB: **Pott's disease**: TB of vertebral bodies
- Tuberculous meningitis
- Tuberculoma: Space occupying lesion in brain
- Gastrointestinal TB: terminal ileum, cecum
- Pericardial TB



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
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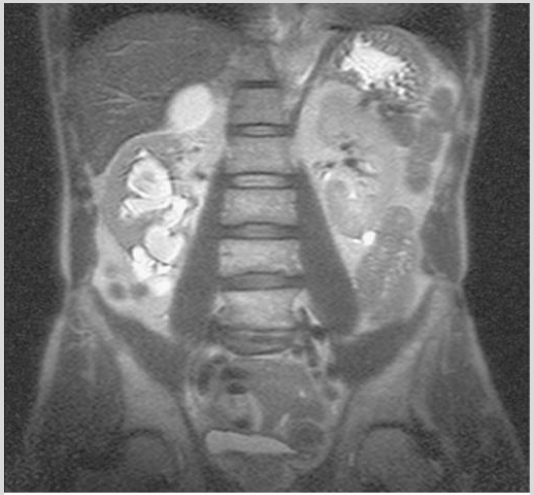
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Clinical Features: Extra-Pulmonary

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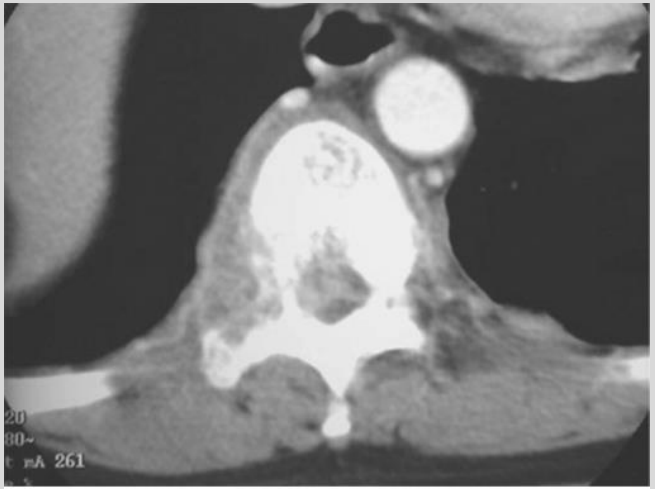


Tuberculoma



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Renal TB



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Pott's Disease

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https://www.cureus.com/articles/78621-diagnostic-and-neurological-overview-of-brain-tuberculomas-a-review-of-literature

Clinical Features: Miliary TB

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Miliary TB

- Due to hematogenous spread of tubercle bacilli throughout the body
- Arise from progressive primary infection or reactivation of latent infection
- Lesions: 1-2mm yellowish granulomas
- Miliary: *looks like millet seeds*
- Liver, bone marrow, spleen, adrenals, meninges, kidneys, fallopian tubes, and epididymis, but can involve any organ



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Image from https://www.ncbi.nlm.nih.gov/books/NBK562300/figure/article-25159.image.f1/?report=objectonly

Diagnosis: Medical History

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- Symptoms
- TB exposures
- Risk factors
- Travel or emigration
- Prior LTBI or TB disease diagnosis
- Prior TST or IGRA results
- Clinical suspicion

https://www.cdc.gov/tb/index.html

Diagnosis: Symptoms

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Symptoms of Pulmonary TB Disease (TB disease usually causes one or more of the symptoms)	**Symptoms of Possible Extrapulmonary TB Disease** (Depends on the part of the body that is affected by the disease)
• Cough (especially if lasting for 3 weeks or longer) with or without sputum production • Coughing up blood (hemoptysis) • Chest pain • Loss of appetite • Unexplained weight loss • Night sweats • Fever • Fatigue	• TB of the kidney may cause blood in the urine • TB meningitis may cause headache or confusion • TB of the spine may cause back pain • TB of the larynx can cause hoarseness • Loss of appetite • Unexplained weight loss • Night sweats • Fever • Fatigue

https://www.cdc.gov/tb/index.html

Diagnosis: Physical Exam

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- General:
  - Pallor, clubbing (advanced disease), generalized wasting
- Pulmonary:
  - May have no abnormalities on examination
  - Consolidation: dullness to percussion, crackles, diminished breath sounds
  - Cavitation: hyperresonance on percussion, amphoric breath sounds
  - Bronchial obstruction: rhonchi
  - Pleural effusion and/or empyema: dullness to percussion, diminished breath sounds
- Extra-pulmonary:
  - Depends on system involved



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Differential Diagnosis

- Pneumonia
- Malignancy
- Non-tuberculous mycobacterium
- Fungal infection
- Sarcoidosis

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Diagnosis: Testing

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Mantoux tuberculin skin test (TST)

- T-cell mediated delayed type hypersensitivity reaction
- Takes 2-8 weeks after initial infection for immune system to react to PPD
- Positive test: Person's body was infected with TB bacteria → need additional tests to determine if LTBI or TB disease
- Negative test: Person's body did not react to test
- False-negative: Anergy, low protein states, diseases affecting lymphoid organs
- False-positive: Infection w/ nontuberculous mycobacteria, previous BCG vaccination
- Absence of reaction to TST does NOT exclude diagnosis



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Diagnosis: Testing

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Mantoux tuberculin skin test (TST)

Tuberculin Reaction Size and Cutoff for Tuberculosis (TB) Preventive Treatment

RISK GROUP	TUBERCULIN REACTION SIZE, mm
HIV-infected persons	≥5
Recent contacts of a patient with TB	≥5[a]
Organ transplant recipients	≥5
Persons with fibrotic lesions consistent with old TB on chest radiography	≥5
Persons who are immunosuppressed—e.g., due to the use of glucocorticoids or tumor necrosis factor α inhibitors	≥5
Persons with high-risk medical conditions[b]	≥5
Recent immigrants (≤5 years) from high-prevalence countries	≥10
Injection drug users	≥10
Mycobacteriology laboratory personnel; residents and employees of high-risk congregate settings[c]	≥10
Children <5 years of age; children and adolescents exposed to adults in high-risk categories	≥10
Low-risk persons[d]	≥15



High-risk medical conditions: silicosis, DM, chronic renal failure, malignancy, gastrectomy, jejunoileal bypass

High-risk congregate settings: correctional facilities, nursing homes, homeless shelters, hospitals & other healthcare facilities

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Diagnosis: Testing

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Interferon Gamma Release Assay (IGRA)

- Measures response of T cells to persisting antigenic signals
- Requires blood draw
- More specific than TST
- Preferred TB test for patients who have received BCG vaccine or have difficult time returning for 2nd appointment to read TST



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Diagnosis: Testing

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TST vs IGRA

TST	IGRA
Tuberculin is injected under the skin and produces a delayed-type hypersensitivity reaction if the person has been infected with *M. tuberculosis*	Blood is drawn for testing; test measures the immune response to the TB bacteria in whole blood
Requires two or more patient visits to conduct the test	Requires one patient visit to conduct the test
Results are available 48 to 72 hours later	Results can be available in 24 hours (depending on the batching of specimens by the laboratory and transport)
Can cause booster phenomenon	Does **not** cause booster phenomenon
Reading by HCW may be subjective	Laboratory test **not** affected by HCW perception or bias
BCG vaccination can cause false-positive result	BCG vaccination does **not** cause false-positive result and infection with most nontuberculous mycobacteria does **not** cause false-positive result
A negative reaction to the test does **not** exclude the diagnosis of LTBI or TB disease	A negative reaction to the test does **not** exclude the diagnosis of LTBI or TB disease

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Diagnosis: Imaging

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Diagnosis: Imaging

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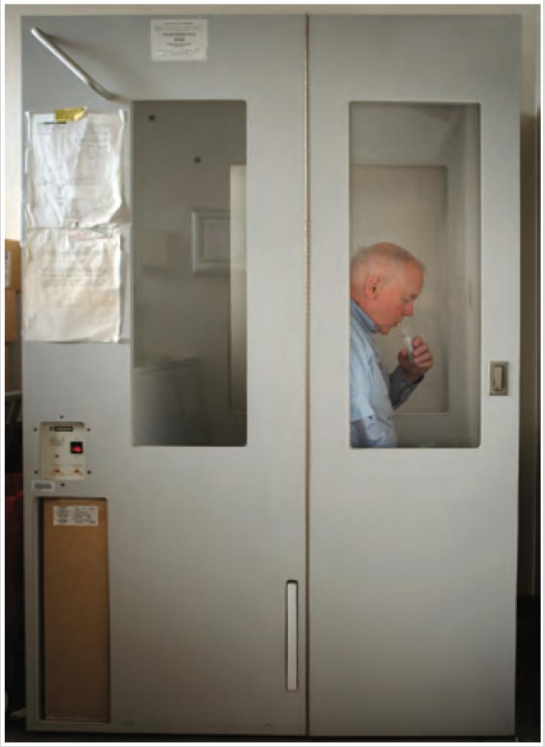
- Posterior-anterior CXR
- Primary:
  - Ghon complex: granuloma in middle/lower lung lobes + hilar lymphadenopathy
- Post-primary:
  - Apical and posterior segments of upper lobes of lung
  - Ill defined patchy consolidation with a cavitary lesion
  - Calcifications
- Miliary:
  - Miliary reticulonodular pattern
- Findings may suggest TB, but **cannot** diagnose TB
- Can appear normal for persons with HIV

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Diagnosis: Sample Collection

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- At least 3 consecutive sputum samples
- Collected in 8-to-24-hour intervals
  - At least one in early morning
- Can be collected via coughing, induction with saline mist, bronchoscopy or gastric aspiration
- Use precautionary infection control measures following sputum induction and bronchoscopy
- Extra-pulmonary TB: urine, CSF, pleural fluid, pus



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Diagnosis: Sample Collection

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Patient Coughing Up Sputum Under Supervision



Performing a Bronchoscopy

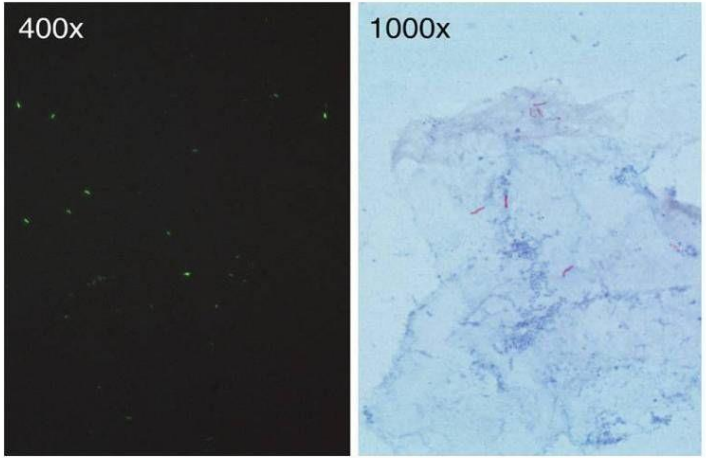
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Diagnosis: Sample

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Acid Fast Bacilli (AFB) Microscopy

- Uses carbolfuchsin method (Ziehl-Neelsen) or fluorescent method (auramine-rhodamine)
- Results available within 24 hours
- Bacilli are counted
  - Greater the number, the more infectious the patient
- Does not confirm a diagnosis of TB because some acid-fast bacilli are not M. tuberculosis
- Negative smears do not exclude TB



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Diagnosis: Sample

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Nucleic Acid Amplification Test

- Detects DNA in samples
- Reliability detect TB in hours as compared to weeks
- Should be preformed on at least one respiratory specimen
- Becoming standard of practice for TB testing
- Does NOT replace need for AFB smear and culture
- Positive NAAT can support diagnosis of TB
- Negative NAAT should not be used as a definitive test to exclude TB



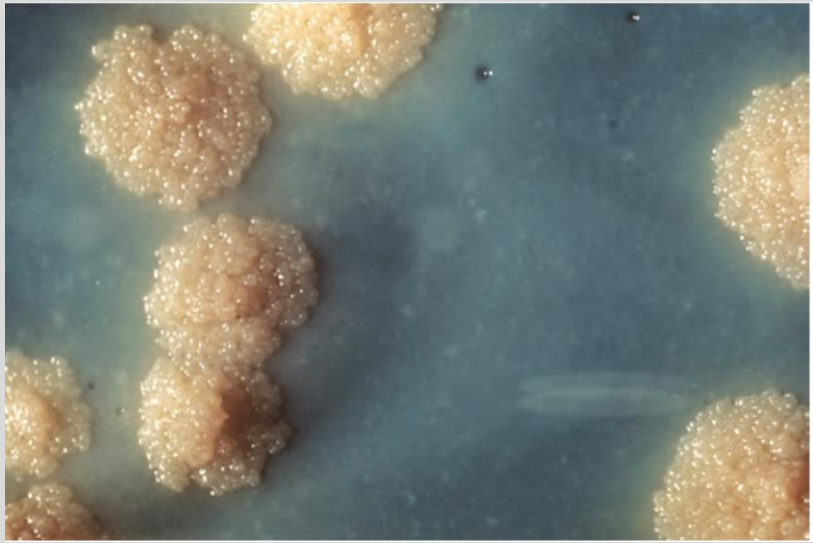
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Diagnosis: Sample

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Mycobacterial Culture

- *GOLD STANDARD FOR CONFIRMATION*
- Culture examinations should be done on all specimens regardless of AFB smear or NAAT results
- Takes 4-6 weeks to incubate on solid media & 10-21 days to incubate on liquid media

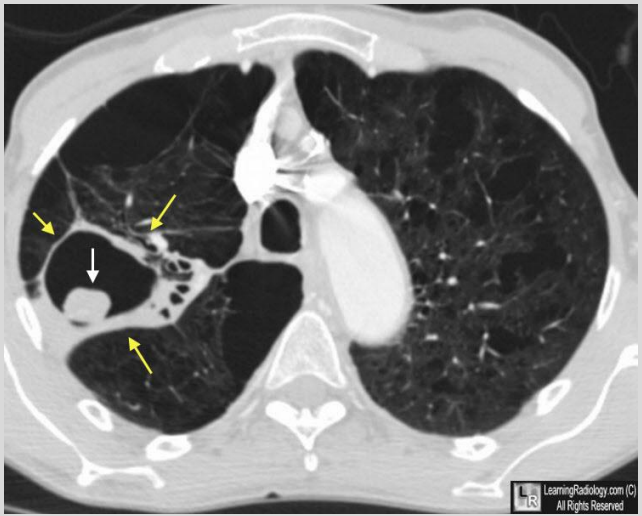


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Post-TB Complications

- Chronic pulmonary disability
- Bronchiectasis
- Lung fibrosis
- Aspergilloma
- Chronic pulmonary aspergillosis

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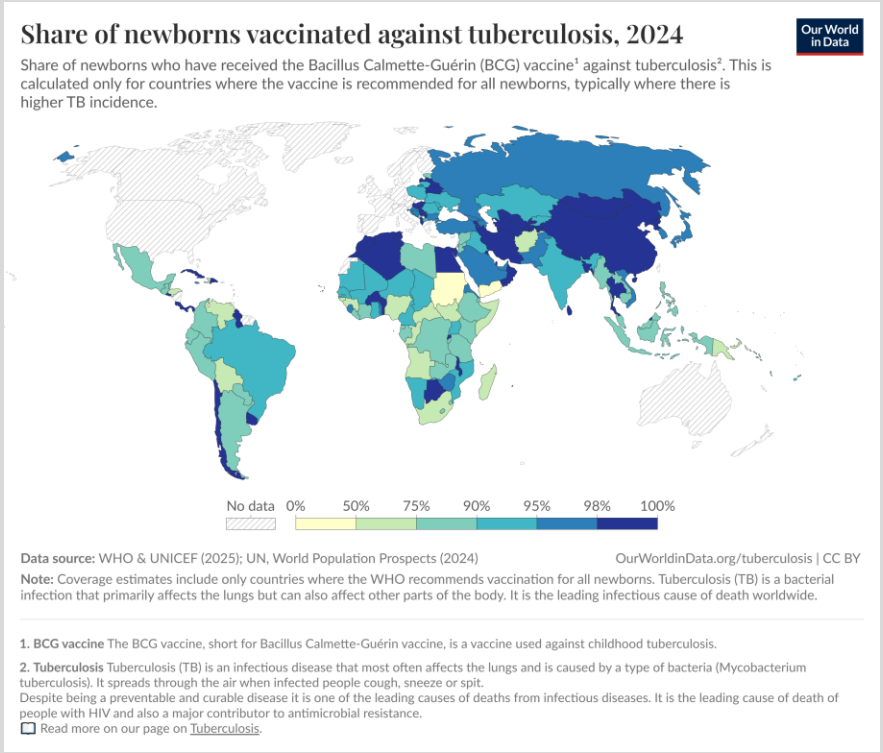
Aspergilloma

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Tobin EH, Tristram D. Tuberculosis. [Updated 2024 Mar 20]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK441916/

Prevention: BCG Vaccination

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- BCG = Bacillus Calmette-Guérin
- Live-attenuated vaccine
- Originally developed from *M. bovis*
- Routine use at birth in countries with high rates of TB
- Not recommended for general use in US
- Induces TST reactivity → use IGRA



Harrison's Principles of Internal Medicine, 21e. Chapter 178: Tuberculosis.

Pai, M., Behr, M. A., Dowdy, D., Dheda, K., Divangahi, M., Boehme, C. C., Ginsberg, A., Swaminathan, S., Spigelman, M., Getahun, H., Menzies, D., & Raviglione, M. (2016). Tuberculosis | Nature Reviews Disease Primers. *Nature Reviews Disease Primer*, *2*, 1–23. https://doi.org/10.1038/nrdp.2016.76

Prevention: Airborne Precautions



- Single occupancy negative pressure isolation room
- PPE: Fit-tested N95
- Limit patient transport and movement
- If patient needs to be transported, patient must wear surgical mask



https://www.cdc.gov/tb/index.html
https://infectioncontrol.ucsfmedicalcenter.org/airborne-isolation-sign

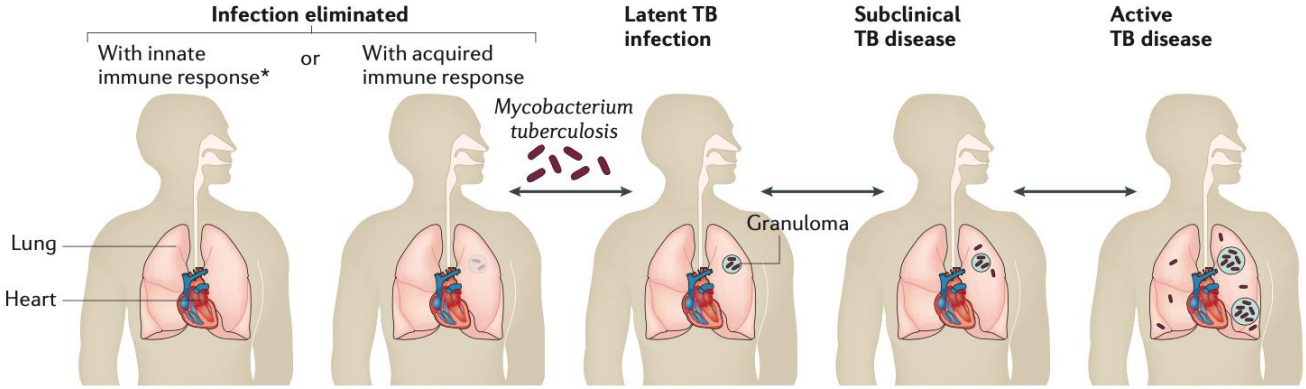
Summary

- TB is a bacterial infection caused by *Mycobacterium tuberculosis.*
- It is spread via inhalation of droplet particles of an infected persons and mainly affects the lungs, but can affect any part of the body.
- Not everyone infected with TB bacteria becomes sick
- There are two TB-related conditions: latent TB infection and TB disease.
- While most patients will not become infectious, if active disease is left untreated, it is deadly.
- The clinical suspicion and diagnostic results are important when determining the patient's course of treatment.

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Summary – Spectrum of TB Disease

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TST	Negative	Positive	Positive	Positive	Usually positive
IGRA	Negative	Positive	Positive	Positive	Usually positive
Culture	Negative	Negative	Negative	Intermittently positive	Positive
Sputum smear	Negative	Negative	Negative	Usually negative	Positive or negative
Infectious	No	No	No	Sporadically	Yes
Symptoms	None	None	None	Mild or none	Mild to severe
Preferred treatment	None	None	Preventive therapy	Multidrug therapy	Multidrug therapy

Source: Pai, M., Behr, M. A., Dowdy, D., Dheda, K., Divangahi, M., Boehme, C. C., Ginsberg, A., Swaminathan, S., Spigelman, M., Getahun, H., Menzies, D., & Raviglione, M. (2016). Tuberculosis | Nature Reviews Disease Primers. *Nature Reviews Disease Primer*, *2*, 1–23. https://doi.org/10.1038/nrdp.2016.76

Core References

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Harrison's Principles of Internal Medicine, 22e. Chapter 183: Tuberculosis. https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3541&sectionid=294819502

Pai, M., Behr, M. A., Dowdy, D., Dheda, K., Divangahi, M., Boehme, C. C., Ginsberg, A., Swaminathan, S., Spigelman, M., Getahun, H., Menzies, D., & Raviglione, M. (2016). Tuberculosis | Nature Reviews Disease Primers. *Nature Reviews Disease Primer*, *2*, 1–23. https://doi.org/10.1038/nrdp.2016.76

Robbins & Cotran Pathologic Basis of Disease, Chapter 8: Infectious Diseases. https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/content/book/3-s2.0-B978032353113900008X

https://www.cdc.gov/tb/hcp/index.html

https://www.cdc.gov/tb/stories/index.html

Knowledge Check

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Question#1: Which of the following groups are more likely to be exposed to or infected with M. tuberculosis?

1. Contacts of people known or presumed to have infectious TB
2. People who live in, or have lived in, high-risk congregate settings (for example, homeless shelters or correctional facilities)
3. Health care workers who serve patients with TB disease
4. People born in or who frequently travel to countries where TB disease is common, including Mexico, the Philippines, Vietnam, India, China, Haiti, and Guatemala

A. 1, 2, and 3 are correct
B. 1 and 3 are correct
C. 2 and 3 are correct
D. Only 4 is correct
E. All are correct

https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html

Knowledge Check

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Question #2: Generally, what percentage of people in the United States who have TB infection and normal immune systems develop TB disease at some point in their lives?

A. 2%
B. 10%
C. 50%
D. 90%

https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html

Knowledge Check

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Question #3: TB blood tests are the preferred method of testing for (select all that apply):

A. Homeless persons or drug users.
B. People who have received the BCG vaccine.
C. Children younger than 5 years of age.
D. A and B
E. All of the above

https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html

Knowledge Check

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Question #4: After TB has been transmitted, how long does it take for the body's immune system to mount a response detectable by a TST or IGRA?

A. 6 months or more
B. 7 to 10 days
C. 2 to 8 weeks
D. 24 hours

https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html

Knowledge Check

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Question #5: When can a TB patient be considered noninfectious?

A. When they are on adequate therapy for two weeks or longer
B. When their symptoms have improved
C. When they have had 3 consecutive negative sputum smear results from specimens collected in 8- to 24-hour intervals (one being an early morning specimen)
D. When all of the above are true

https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html

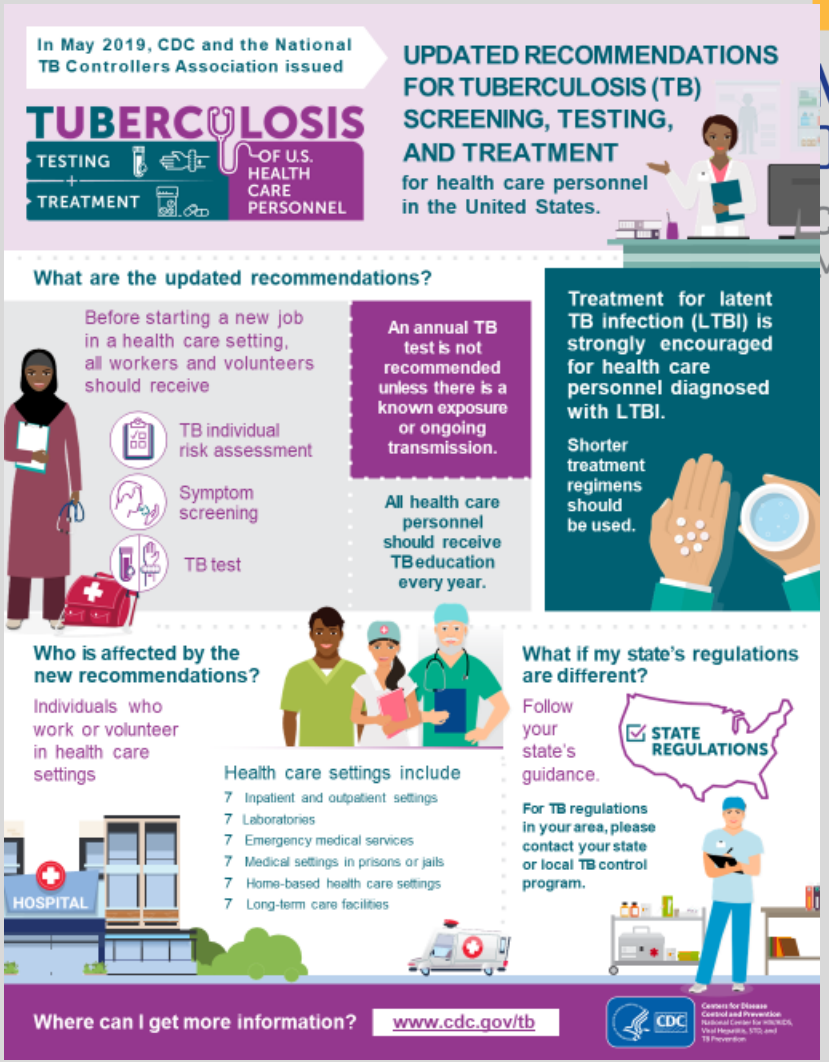
Knowledge Check

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Question #6: Which of the following confirms the diagnosis of TB?

A. Chest x-ray
B. Detection of acid-fast bacilli in smear examination
C. Positive tuberculin skin test
D. Positive culture of M. tuberculosis

https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html



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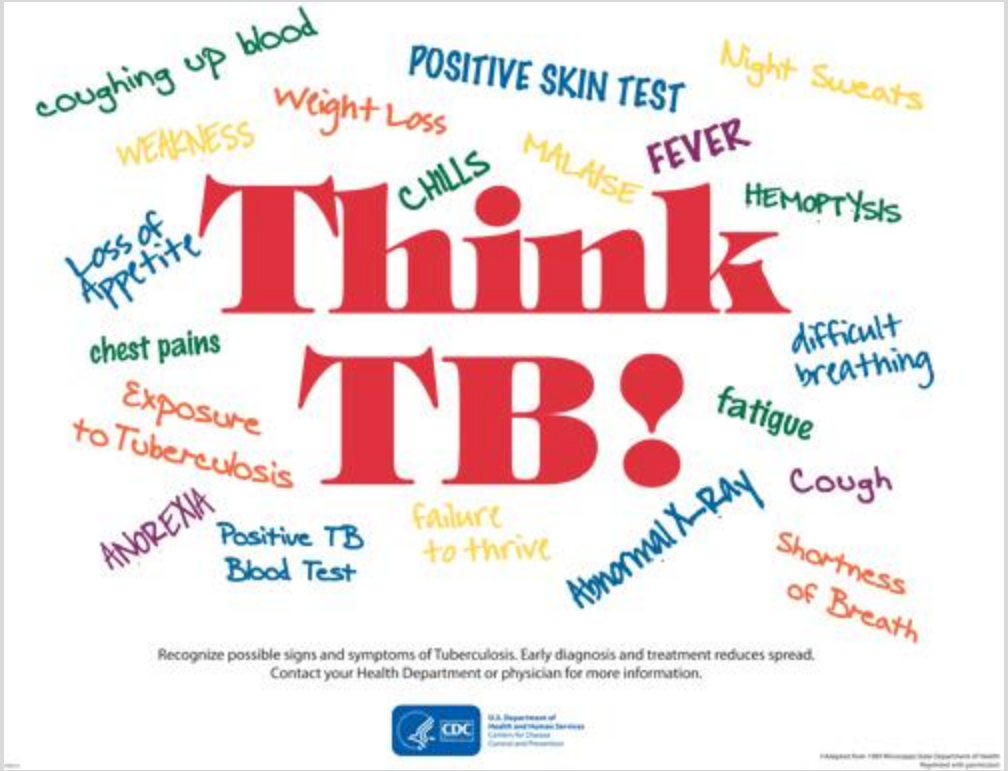
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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R



Knowledge Check Answer Key

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1. E
2. B
3. D
4. C
5. D
6. D

https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html